

Z-basis metrics for parameterized quantum circuits: exact identities, structural blind spots, pole-damped optimization, and a relaxation-aware entropy benchmark

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Abstract

The Qang (qg) framework [1] expresses single-qubit rotations through two measurement-level quantities, the polar bias $qg_Z = \langle \sigma_z \rangle$ and the outcome entropy $qg_S = H(p_0)$. We report results obtained with the open-source reference implementation **qang**. First, the two metrics are linked by an exact, branch-free identity, $qg_S = H((1 + qg_Z)/2)$, which holds for any single-qubit state. Consequently, any new multi-qubit information must come from the joint outcome distribution. We name the resulting Z-basis total correlation and give its bounds and its scope: it is not an entanglement measure. Second, we identify exact blind spots: graph-state entanglement and readout-only dephasing are invisible to every Z-diagonal statistic, and qg-space updates cannot reach optima outside $[0, \pi]$ because of the arccos range. On H_2 they recover none of the correlation energy. Third, we study a pole-damped gradient step whose damping schedule is $|dqg_Z/d\theta|$. On a 300-landscape benchmark, H_2 , and a 4-qubit LiH Hamiltonian with a mixed R_y/R_x ansatz, it trades 3–20 $\times$ more iterations at well-tuned learning rates for convergence where plain gradient descent diverges. Fourth, as a benchmark, qg_S tracks heavy output probability ($r = -0.93$) and needs no calibration, unlike linear XEB. However, it is *not* monotonic under amplitude damping: under device-calibrated noise with added idle relaxation it falls below its own noiseless value. Pairing qg_S with the register-averaged qg_Z removes the ambiguity in 23 of 24 circuits tested. A summary section lists thirty-four follow-up studies (QML encoding, control quantization, noise-type detection, circuit knitting, chemistry, error mitigation, thermal states, error correction, Hubbard dynamics, constrained QAOA, qubit characterization, syndrome tracking, coherence, few-shot estimation, IonQ noise models, shot-noise propagation, neural quantum states, coherent gate errors and symmetry checks in rotated bases), each with its classical or standard control; the symmetry-witness results are developed in a companion paper. Every number is reproduced by a script and pinned by a regression test in the public repository.

1 Introduction

Parameterized quantum circuits are written in angles, but everything a device returns is a distribution over measurement outcomes. The Qang framework [1] makes that second view the primary one: for a qubit at polar angle θ ,

$$qg_Z(\theta) = \cos \theta = \langle \sigma_z \rangle \in [-1, 1], \quad qg_S(\theta) = H(\cos^2 \frac{\theta}{2}) \in [0, 1], \quad (1)$$

with $H(p) = -p \log_2 p - (1-p) \log_2 (1-p)$. This paper collects results obtained with the reference implementation **qang**, with the aim of stating where the metrics are exact, where they are blind, and where they mislead. Section 2 gives exact identities, Section 3 the structural blind spots,

Section 4 a pole-damped optimizer on molecular Hamiltonians, and Section 5 the behavior of qg_S as a benchmark, including a device-calibrated validation. Section 6 summarizes later results, and Section 7 lists the limitations together.

All results are produced by scripts in the `examples/` folder of the repository and pinned by regression tests (Section 8). Molecular Hamiltonians were derived once with PySCF [2] and are hard-coded, so neither PySCF nor qiskit-nature is a dependency.

2 Exact identities

2.1 A branch-free link between qg_Z and qg_S

Inverting $qg_S \mapsto \theta$ requires choosing a half-domain, because $qg_S(\theta) = qg_S(\pi - \theta)$. The opposite direction needs no choice.

Proposition 1. *For any single-qubit density matrix ρ , with $qg_Z = \text{Tr}(\rho\sigma_z)$ and $qg_S = H(\langle 0|\rho|0\rangle)$,*

$$qg_S = H\left(\frac{1 + qg_Z}{2}\right), \quad \frac{d qg_S}{d qg_Z} = \frac{1}{2} \log_2 \frac{1 - qg_Z}{1 + qg_Z}. \quad (2)$$

Proof. $qg_Z = p_0 - p_1 = 2p_0 - 1$ with $p_0 = \langle 0|\rho|0\rangle$; substitute into $H(p_0)$ and differentiate. $\square$

The proof uses only the Z-diagonal population. The identity therefore holds however the state was prepared (R_y , R_x , a noisy channel, or a partial trace of a larger register), and the tests check it on random Haar states. The derivative diverges at $qg_Z = \pm 1$. This is the entropy-side counterpart of the Jacobian singularity $d\theta/dqg_Z = -1/\sin\theta$.

2.2 Z-basis total correlation

Applied to each reduced single-qubit state, Proposition 1 makes the per-qubit qg_Z profile and the per-qubit marginal qg_S two encodings of the same information. Anything new about an n -qubit register must come from the joint distribution of the Z-basis outcomes $X_1, \dots, X_n$. We define

$$C_Z(\rho) = \sum_{i=1}^n H(X_i) - H(X_1, \dots, X_n), \quad (3)$$

Watanabe’s total correlation [3] of the outcomes. For $n = 2$ it equals the mutual information. By subadditivity, $0 \leq C_Z \leq n - 1$ bits, with $C_Z = 0$ iff the outcomes are independent. Product states give 0, Bell states 1 and GHZ $_n$ states $n - 1$ (the maximum).

Remark 1 (Scope). C_Z is a classical correlation measure and is not a witness of entanglement in either direction. The separable mixture $\frac{1}{2}(|00\rangle\langle 00| + |11\rangle\langle 11|)$ has $C_Z = 1$, as does a Bell state. Every graph state has $C_Z = 0$ (Section 3).

3 Structural blind spots

Graph states. A graph state [4] is $H^{\otimes n}$ followed by CZ gates. $H^{\otimes n}$ makes every $|\text{amplitude}|^2$ equal to 2^{-n} , and CZ is diagonal. For every graph, therefore, each qubit has $qg_Z = 0$, the joint qg_S is maximal, and $C_Z = 0$. The entanglement, certified independently through the stabilizers $X_i \prod_{j \in N(i)} Z_j$, lives entirely in phases.

Readout dephasing. Dephasing preserves the diagonal of ρ . Applied only before measurement, it therefore leaves qg_S exactly unchanged at every strength, including full dephasing (checked to machine precision on a 4-qubit Quantum Volume circuit). The same channel applied *inside* the circuit is visible: later entangling gates turn the lost coherence into randomized populations, and qg_S rises monotonically.

Table 1: Success rates over 300 random one-qubit landscapes per learning rate, started near a pole.

η	0.3	1.0	2.0	3.0	5.0	10.0
plain success	0.997	0.773	0.187	0.087	0.057	0.027
damped success	0.963	1.000	0.477	0.357	0.253	0.130
damped trapped	0.017	0.000	0.030	0.023	0.040	0.067

The arccos range. A qg-space parameter update is mapped back through $\theta = \arccos(qg_Z) \in [0, \pi]$. On the H_2 Hamiltonian (0.735 Å, STO-3G, two qubits after parity reduction), the single-excitation optimum lies at $\theta \approx -0.22$. Starting from Hartree–Fock, raw, clipped and Tikhonov-regularized qg-space descent all stop at exactly the Hartree–Fock energy (-1.1169990 Ha, error 2.0×10^{-2} Ha). They recover none of the correlation energy, whereas θ -space descent reaches the FCI energy to 10^{-9} Ha. Regularizing the Jacobian cannot fix this, because the limit comes from the range of arccos, not from its derivative.

4 Pole-damped gradient descent

To keep updates in θ -space while still using the pole geometry, we damp each parameter’s step by $|dqg_Z/d\theta_i|$, floored at ε :

$$\theta_i \leftarrow \theta_i - \eta \max(|\sin \theta_i|, \varepsilon) \frac{\partial E}{\partial \theta_i}. \quad (4)$$

This is position-dependent trust-region damping in the Levenberg–Marquardt tradition [5, 6]. It is not a new class of optimizer; the only framework-specific element is that the schedule is derived from the geometry rather than tuned. Because the factor depends only on populations, and $R_x(\theta)|0\rangle$ and $R_y(\theta)|0\rangle$ have identical populations, it applies unchanged to either axis. We use $\varepsilon = 0.05$ throughout; results are insensitive to this choice.

Random landscapes. For 300 random landscapes $E = h_z \cos \theta + h_x \sin \theta$ per learning rate, started near a pole (Table 1), damping costs a few percent of success at a safe rate. At $\eta = 1$ it succeeds in 100% of cases versus 77%, and at larger rates it succeeds 2.5–5 times more often. The cost is a trapping mode, in which the optimizer oscillates at its starting pole; it reaches 6.7% at $\eta = 10$.

H_2 , two coupled parameters. Ansatz $R_y(\theta_0) \otimes R_y(\theta_1)$ followed by CX; the optimum has $\theta_0 = \pi$ (a pole). From $(1.0, 10^{-6})$, damping needs about $3\times$ more steps to reach chemical accuracy at safe rates (603 vs. 196 at $\eta = 0.05$). At $\eta = 2.5, 3$ and 4, plain descent never reaches chemical accuracy, while the damped version reaches the FCI energy in 10, 8 and 6 steps.

LiH, mixed axes. LiH at 1.5459 Å (STO-3G, 2 electrons in 3 spatial orbitals, parity mapping) gives 4 qubits and 52 Pauli terms, the scale of early hardware VQE work [7, 8]. The ansatz is $R_y(\theta_0)$, $R_y(\theta_1)$ on the occupied orbitals and $R_y(\theta_2)$, $R_x(\theta_3)$ on the virtual ones, followed by CX($2 \rightarrow 0$) and CX($3 \rightarrow 1$). At $(\pi, \pi, 0, 0)$ its state has overlap 1 with qiskit-nature’s Hartree–Fock circuit. All four optimal parameters lie within 0.04 rad of a pole, which makes the cost larger than for H_2 : 355 vs. 17 steps at $\eta = 0.3$. The benefit is also larger (Table 2). The R_x phase is physical: after the CX it changes the energy relative to R_y at the same angle, although the single-qubit populations are identical.

Table 2: LiH, start $(\pi, \pi, 10^{-6}, 10^{-6})$: steps to sustained convergence (10^{-5} Ha) to the ansatz optimum -0.0440152 Ha (active-space electronic energy; HF -0.0437813 , FCI -0.0448309).

η	0.3	1.0	2.0	2.5	4.0	7.0	10.0
plain θ	17	5	never	never	never	never	never
pole-damped	355	106	53	42	26	15	10

5 qg_S as a benchmark

Heavy output probability and XEB. On 10 random 4-qubit Quantum Volume circuits [9] mixed with global depolarizing noise, qg_S and heavy output probability have Pearson $r = -0.926$ over 110 points. Linear XEB [10] has a noiseless value $A - 1$, with $A = 2^n \sum_x p(x)^2$ ranging from 1.23 to 2.33 on the same circuits family ($n = 3-6$), so it needs circuit-specific calibration; qg_S does not. Conversely, XEB’s estimator is a sample mean and is unbiased at any shot count, while the plug-in entropy is biased low (bias -0.054 at 50 shots on 4 qubits). The Miller-Madow correction [11] reduces that bias about fivefold.

Amplitude damping breaks monotonicity. The reading “more noise $\Rightarrow$ higher qg_S ” holds only for channels whose fixed point is maximally mixed. With amplitude damping applied after every two-qubit block (Fig. 1a), qg_S rises from 0.919 to 0.983 at $\gamma = 0.1$ and then falls to 0 at $\gamma = 1$. At $\gamma = 0.4$ it reads 0.918, almost the noiseless value.

The fix: register-averaged qg_Z . The mean $q\bar{g}_Z = \frac{1}{n} \sum_i \text{Tr}(\rho_i \sigma_z)$ is pulled towards 0 by unital noise and towards +1 by relaxation. At $\gamma = 0$ and 0.4 the pairs $(qg_S, q\bar{g}_Z)$ are (0.919, -0.029) and (0.918, $+0.310$). $q\bar{g}_Z$ rose monotonically along a 21-point γ grid in 23 of 24 circuits ($n = 3, 4, 5$; seeds 0–7); the exception dips by 0.015 before rising. $q\bar{g}_Z$ is linear in ρ , so its estimator from counts is an exact sample mean (Appendix A).

Device-calibrated validation. We ran the same circuit through Qiskit Runtime on the `fake_brisbane` backend, a simulator whose noise model is built from a real IBM device’s calibration data. An idle delay before readout served as a T1 knob (Fig. 1b). Heavy output probability fell monotonically (0.685 to 0.448) and $q\bar{g}_Z$ rose monotonically (-0.036 to $+0.370$). qg_S rose to 0.977 at 50 μs and then fell to 0.892 at 200 μs , below its noiseless value of 0.919. Read on its own, qg_S would rank the most relaxed operating point as the least noisy.

The same script evaluated the LiH ansatz without error mitigation. The raw error was $+0.036$ Ha at Hartree-Fock and $+0.035$ Ha at the optimum. This is about $20\times$ chemical accuracy and about $150\times$ the 0.23 mHa difference the optimizer resolves noiselessly. A hardware run of this VQE therefore needs mitigation first.

Randomized benchmarking and ZNE. Under per-gate depolarizing noise p , the RB survival signal is exactly $qg_Z(m) = (1 - p)^{m+1}$, identical across random Clifford sequences to 10 digits in our runs [12]. For zero-noise extrapolation [13], extrapolation in qg_Z -space is exact for Bloch-vector shrinkage but biased (1.8×10^{-3}) for coherent angle drift; θ -space shows the reverse. One should extrapolate in whichever space makes the noise linear. In barren-plateau settings [14, 15], regularized qg -space gradients keep the exponential variance decay, and raw qg -space gradients still diverge at the poles.

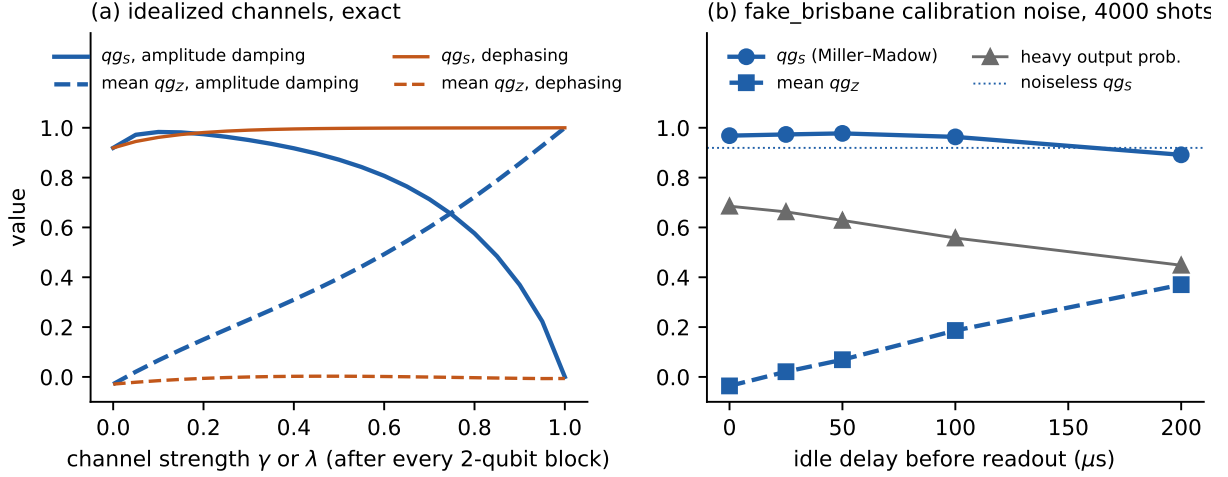


Figure 1: (a) 4-qubit Quantum Volume circuit, exact density-matrix simulation with amplitude damping or dephasing after every two-qubit block. qgs is non-monotonic under amplitude damping, while the mean qgz separates the regimes. (b) The same circuit under **fake_brisbane** calibration noise with an idle delay before readout. Heavy output probability and mean qgz are monotonic; qgs falls below its noiseless value (dotted).

6 Further results (summary)

After the first draft, the same code base was used for thirty-four further studies. Each compares a qg-based choice with the standard one and with a classical control, and each is documented in `RESEARCH_NOTES.md` (section numbers below) with a script and pinned tests. Table 3 lists them. The column “gain” states against what the qg choice wins, if at all. None of them is a quantum advantage over classical computation. Where a classical method was run, it matched or beat the quantum model at these sizes. The symmetry-witness studies (§20, §21, §24, §28–§33, §38, §39, §48, §50–§55) are developed in a companion paper [16].

7 Limitations

(i) The Jacobian singularity at $\theta \in \{0, \pi\}$; regularization costs exactness there. (ii) The arccos range: qg-space updates cannot reach $\theta < 0$. (iii) Z-basis blindness to phase-encoded entanglement and to readout dephasing. (iv) C_Z is not an entanglement measure. (v) qgs is not monotonic under T1. The pair $(qgs, q\bar{q}z)$ fixes this empirically, in 23 of 24 circuits, not provably. (vi) Finite-shot bias of qgs . (vii) Pole damping costs iterations and has a trapping mode. (viii) All device results, here and in the follow-up studies, are calibration-based or vendor-model simulations. Runs on real hardware (`-mode ibm`, IonQ) are prepared but have not yet been performed.

8 Code availability

The library, every example, and the regression tests pinning every number above (973 tests at the time of this revision) are available at <https://github.com/Viny2030/qang> (MIT license). Figure 1 is regenerated by `manuscript/make_figures.py`. Derivations with additional detail are in `RESEARCH_NOTES.md`.

Acknowledgments and use of AI tools

Generative AI tools (Anthropic Claude 3.5 Sonnet and Claude Opus 5.5; OpenAI GPT-4o) assisted with code generation and refactoring, test expansion, numerical experiments, and drafting of this text. The author framed the research questions, made the design decisions, and reviewed, edited and validated all AI-assisted output, including every derivation and number reported here.

A Riesz–Fréchet and why qg_Z is the canonical linear readout

Operators on $\mathbb{C}^{2^n}$ form a Hilbert space under $\langle A, B \rangle = \text{Tr}(A^\dagger B)$. By the Riesz–Fréchet representation theorem [22, 23], every linear functional on it has the form $\rho \mapsto \text{Tr}(O\rho)$ for a unique O . In finite dimension this is elementary linear algebra, so the appendix records a foundation rather than a new result. The functional $\rho \mapsto qg_Z(\rho)$ is represented by σ_z , and $q\bar{g}_Z$ by $\frac{1}{n} \sum_i Z_i$. Every linear readout, including linear XEB, has this form and hence an unbiased sample-mean estimator. qg_S is strictly concave in ρ , so no operator represents it, and its plug-in estimator is biased by Jensen’s inequality. By Proposition 1, qg_S is a fixed scalar function of the linear quantity qg_Z , so it carries no information about a single qubit beyond what σ_z extracts. (Riesz’s 1918 lemma on almost-orthogonal vectors concerns infinite-dimensional spaces and plays no role here.)

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Table 3: Follow-up studies (section numbers of RESEARCH_NOTES.md).

§	topic	main number	gain
15	natural gradient, cut cost, few-shot qg_z	$QFI = 1/(1 - qg_z^2)$; cut $\gamma = 1 + 2\sqrt{1 - qg_z^2}$; Haar prior 92% vs 8% coverage near a pole	closed forms; vs delta method
16, 19	QML encoding $\theta = \arccos x$	Chebyshev models fit polynomials to 10^{-13} vs 5×10^{-5} ; $\sim 10\times$ with shots	vs angle encoding [17]; classical Chebyshev fit wins
17	control grid uniform in qg_z	probability error $\pi/2$ smaller; distribution loading TVD lower in 43/45 cases	vs θ grid; loses on fidelity near poles
18	T1 vs unital noise from counts	accuracy 0.88–0.91 vs 0.70–0.76 (XEB, HOP), strong noise only	vs XEB/HOP
20	IonQ noise models	$H_2 + qg_z$ filter ~ 12 vs ~ 30 mHa, 2–3% shots dropped	vs readout mitigation
21	H_2 electron-number witness	5.5 vs 18 mHa (HF 20.3); $11\times$ below HF at 2.5 Å; no gain on 6-qubit LiH (see §50–54)	vs readout mitigation, HF
22	ODE solver [18]	exponential decay up to $17\times$ better; not in general	vs angle encoding; spectral method wins
23	local cost $(1 - q\bar{g}_z)/2$	gradient variance $5600\times$ larger at 12 qubits; light cone trains 100 qubits classically	vs global cost [15]; classically simulable [19]
24	qg_z filter vs ZNE by noise type	witness predicts when the filter works; ZNE + filter 2.0 mHa on fake_brisbane	complementary to ZNE
25	thermal qubit, $qg_z = \tanh \beta h$	optimal thermometer $qg_z \operatorname{artanh} qg_z = 1$ [20]	none (prior, not qg)
26	which gates to cut	exact optimum over 3795 partitions; $2\times$ fewer shots than counting	vs angle-blind rules; same as find_cuts
27	repetition code choice under T1	$qg_z + qg_x$ policy regret 2.7% vs 46–258%	decision rule
28	spin-resolved filters	23.2 $\rightarrow$ 14.7 mHa under depolarizing noise; none on fake_brisbane	conditional
29	1D Hubbard, 8 qubits	ZNE + spin filters 6–9 $\times$ below raw; wall at ≥ 188 ECR	vs raw, ZNE
30	QAOA, exactly K	filter doubles $P(\text{opt})$ of XY-QAOA; penalty QAOA $\approx$ random guess	vs QAOA; greedy wins
31	qubit characterization	unbiased T_{eff} (± 1 mK) and readout error, where the standard suite reads 0 K	vs standard calibration
32	Leung code vs T1	infidelity $\approx 0.92\gamma^2$ vs $\gamma/3$; rule: Leung iff $T_2 > T_1$	decision rule; code known
33	syndromes as witnesses	$\langle XXXX \rangle = qg_x^4$; drift tracking 13% regret with no probe shots	vs periodic recalibration
34	entropy gap = coherence	qg sum bound captures 92–100% of Z coherence (thermal Ising)	none on graph states
35	few-shot tomography	per-axis Haar prior best closed form on mixed states	loses on pure states; prior, not qg
36	Ramsey operating point	$F = (V^2 - qg_z^2)/(1 - qg_z^2)$	none (known rule)
37	Grover as $T_{2k+1}(qg_z)$	$F_m = m^2/(1 - qg_z^2)$; MLAE $\sim 9\times$ below Monte Carlo	none (known); noise must be modelled
38	IonQ noise models	filter doubles XY-QAOA; spin filters pay at depth	vs QAOA; greedy wins
39	native-gate ZNE on IonQ	CX folding is undone; at 2000 shots filter beats ZNE in RMS	depends on shot budget
45	shot noise, mixed states and registers	$\text{Var}(\hat{\theta}) = [1 + (1 - r^2)/(r^2 \sin^2 \theta)]/N$; witness variance 0 on Dicke states, $n\times$ on GHZ	none (standard statistics); corrects §5 scope
47	RBM $I-\eta$ plane (review [21])	bounds are the marginal extremes: LB at $qg_z^v = qg_z^h = 0$; field h_z moves trained RBMs off LB	explains a claimed learning principle by Z_2 symmetry
48	coherent MS over-rotation	native folding blind to it (ZNE + 2.96 mHa at 5%); filter removes 94%	ZNE + filter most robust
50	filter reach (H_2 vs LiH)	LiH error in X/Y groups (26–166 mHa), not routing; 8/9th-fraction rescaling overcorrects $2-3\times$	negative result
51	error bars for qgs	Wilson interval for qg_z mapped through $qgs = H((1 + qg_z)/2)$ covers 0.90–0.99; Wald bootstrap fail	vs Wald, bootstrap, Haar-Bayes